

Project Progression

June 2026

MR ROKESH TECHNOLOGY

Cross-Track Project Progression (L1–L5)

Standard project difficulty ladder applied across all technical tracks. See individual track files for track-specific project names and capstone specs.

Project Levels

Level	Name	Complexity	When	Example Projects
L1	Mini	Single-feature, no backend	2 weeks – 1 month	Calculator, To-Do, Landing page
L2	Guided	API or DB integration, mentor-led	1 – 3 months	Weather app, Blog CMS, Expense tracker
L3	Integrated	Multi-module, auth, real deployment	3 – 6 months	E-commerce, Job portal, Chat app
L4	Capstone	Production-grade, full stack + deploy	6 months	Full production app with auth, payments, deploy
L5	Advanced	Multi-service, cloud-native, AI-integrated	9 months	SaaS platform, ML microservice, enterprise dashboard

L1 — Mini Projects

Characteristics: - 1–3 days of focused work - Single page or single-feature app - No authentication required - Runs locally or static deploy

Acceptance Criteria: - [] Core feature works as demonstrated - [] Code pushed to GitHub with README - [] Basic styling applied (not raw HTML) - [] Student can explain code in 5-minute demo

Examples by Track:

Track	L1 Examples
Full Stack	Personal landing page, To-Do app
Database	Library schema design + basic queries
Programming	Calculator CLI, 20 basic programs
Mobile	Hello World + Calculator app
AI/ML	Data cleaning notebook
Cybersecurity	Security awareness quiz + Kali setup
DevOps	Linux admin tasks script
Cloud	EC2 instance + S3 static site
UI/UX	Mood board + 3 mobile screens
Office/Tally	Formatted business report
Digital Marketing	Persona document + SEO audit

L2 — Guided Projects

Characteristics: - 1–2 weeks of work - External API or database integration - Mentor provides architecture guidance - Deployed to free tier (Render, Vercel, etc.)

Acceptance Criteria: - ☐ All CRUD operations or API flows work - ☐ Error handling for common failure cases - ☐ README with setup steps and screenshots - ☐ Live demo URL included - ☐ At least 3 meaningful Git commits

Examples by Track:

Track	L2 Examples
Full Stack	Weather app, Blog frontend
Database	Student management DB, Blog NoSQL DB
Programming	Expense tracker CLI, 50 LC Easy problems
Mobile	Weather/News app, Expense tracker
AI/ML	Churn prediction model, EDA report
Cybersecurity	Footprinting report, DVWA labs
DevOps	CI pipeline, 3-tier containerized app
Cloud	Scalable web app on AWS
UI/UX	App prototype (mid-fi), UI kit
Office/Tally	Financial tracker in Excel
Digital Marketing	30-day content plan, ad campaign

L3 — Integrated Projects

Characteristics: - 2–4 weeks of work - Frontend + backend + database - User authentication - Production deployment with custom domain (optional)

Acceptance Criteria: - ☐ Auth flow (register, login, logout) works - ☐ Role-based or user-scoped data access - ☐ Responsive UI on mobile and desktop - ☐ Environment variables for secrets (no hardcoded keys) - ☐ API documentation or Postman collection - ☐ 5-minute demo video recorded

Examples by Track:

Track	L3 Examples
Full Stack	Blog CMS (MERN), Task manager API
Database	Multi-tenant SaaS DB design
Programming	Custom data structure library
Mobile	Food ordering app, Login sync app
AI/ML	Resume classifier, Sentiment bot
Cybersecurity	Full pentest report on test app
DevOps	Containerized microservices on K8s
Cloud	Multi-cloud DR solution (partial)
UI/UX	End-to-end UX case study + HTML build
Digital Marketing	Full funnel campaign for local business
Office/Tally	Full company books in Tally

L4 — Capstone Projects

Characteristics: - 4–6 weeks of work - Production-grade architecture - Payments, admin panel, or advanced feature - CI/CD pipeline - Comprehensive documentation

Acceptance Criteria: - ☐ All L3 criteria met - ☐ Payment gateway OR admin dashboard OR real-time feature - ☐ Automated tests for critical paths (minimum 5 tests) - ☐ CI/CD deploys on push to main branch - ☐ Security: HTTPS, input validation, auth tokens - ☐ Case study document (problem, solution, architecture, results) - ☐ Mock interview presentation prepared

Capstone Template — Evaluation Rubric:

Criterion	Points	Description
Functionality	30	All features work; edge cases handled
Code Quality	20	Clean structure, naming, no dead code
Architecture	15	Sensible separation of concerns
Deployment	15	Live, stable, env vars configured
Documentation	10	README, API docs, architecture diagram
Demo	10	Clear 10-min presentation + Q&A
Total	100	Pass: 60+

Examples by Track:

Track	L4 Capstone Examples
Full Stack	E-commerce, Job Portal, Social Media Clone
Database	Hospital Management / ERP hybrid schema
Programming	Splitwise clone, Pathfinding engine
Mobile	E-commerce or Food Delivery app
AI/ML	End-to-end ML pipeline with live deployment
Cybersecurity	End-to-end pentest + professional report
DevOps	Production-grade observable system
Cloud	Multi-cloud disaster recovery solution
UI/UX	Real client project with case study
Digital Marketing	GA4 + Google Ads certification portfolio
Office/Tally	Live client accounting project

L5 — Advanced Capstones

Characteristics: - 6–8 weeks each (2 capstones in 9-month track) - Multi-service or cloud-native architecture - AI/ML integration OR enterprise-scale features - Open-source contribution component

Acceptance Criteria: - ☐ All L4 criteria met - ☐ Two or more services/modules with clear boundaries - ☐ Caching, queues, or real-time features where applicable - ☐ Monitoring/logging in production - ☐ Performance tested under load (basic) - ☐ Open-source PR or documented contribution - ☐ System design diagram and trade-off documentation

Examples by Track:

Track	L5 Capstone Examples
Full Stack	SaaS product + Enterprise RBAC dashboard
Database	Analytics platform + Real-time chat backend
Programming	Algorithm library + Production-grade application
Mobile	Marketplace app + AI productivity app
AI/ML	Multi-modal AI product + Production ML microservice
Cybersecurity	Enterprise network pentest + Blue team IR simulation
DevOps	GitOps platform + Self-healing multi-service architecture
Cloud	Enterprise multi-tier architecture + Serverless event platform
UI/UX	SaaS product design + Enterprise design system
Digital Marketing	Startup launch campaign + Agency portfolio (3 clients)

Project Timeline by Duration

Duration	Project Levels Covered
2 Weeks	L1 only
1 Month	L1 + L2 (start)
3 Months	L1 + L2 + L3
6 Months	L1 – L4
9 Months	L1 – L5

Related Documents

- [Duration Framework](#)
- [Combo Packages](#)
- [Master Syllabus](#)
- [Track Index](#)